



25CY205 DATABASE MANAGEMENT SYSTEMS

Hours Per Week :

L	T	P	SL	C
2	2	2	2	4

PREREQUISITE KNOWLEDGE: Knowledge of basic Discrete Mathematics / Discrete Structures like - Sets, relations, functions, Propositional logic, Boolean algebra and logic (important for query optimization and relational algebra).

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the core principles and techniques of Database Management Systems (DBMS), with a focus on Relational Database Management Systems (RDBMS) widely used in business, scientific, and engineering applications. It covers key topics including ER modeling, relational algebra, SQL, normalization, and the relational model to support effective database design. The course also addresses essential system concepts such as transaction processing, concurrency control, and recovery. The objective is to provide students with the theoretical foundation and practical skills needed to design, query, and manage relational databases using modern tools and techniques.

MODULE-1

UNIT-I

6L+6T+6SL+6P=24 Hours

DATABASE SYSTEM CONCEPTS:

Databases And Database Users: Introduction and Characteristics; Advantages of DBMS, and database users.

Database System Concepts and Architecture: Data models, Schemas and instances; Three-Schema architecture and data Independence; Database languages and interfaces; The database system environment; Centralized and Client-Server architectures for DBMS.

Conceptual Data Modelling and Database Design: Entity types, Entity sets, Attributes and keys; Relationship types, Relationship sets, Roles and structural constraints; Weak entity types; Relationship types.

UNIT-II

6L+6T+6SL+6P=24 Hours

RELATIONAL DATABASE DESIGN:

Relational Database Design by ER-To-Relational Mapping: Relational Database design using ER-to-Relational mapping.

The Relational Data Model and Relational Database Constraints: Relational model concepts; Relational model constraints and Relational database schemas.

Relational Algebra: Unary relational operations - SELECT and PROJECT; Relational algebra operations from set theory; Binary relational operations - JOINS.

SQL: SQL data definition and data types; specifying constraints in SQL, Basic retrieval queries in SQL; INSERT, DELETE, and UPDATE statements in SQL

PRACTICES:

- Design an ER diagram for an Airline Reservation System, identifying entity types, attributes, and relationships.
- Construct an ER model for a Car Rental System, including weak entities and constraints.
- Convert the ER diagram of a Student-Course Enrolment System into a relational schema.
- Development of Relational Database schemas for Sailors/Company/Universities using DDL.
- Apply various DML Commands such as select, insert, update etc. of SQL on Relational Database.
- Design of Relational Database schemas by specifying different types of Constraints.
- Apply various Relational Database operators (Arithmetic, Logical & comparison) and string-

matching constructs of SQL.

- Expressing queries using Aggregate Functions of SQL on Relational Database.
- Implement PRIMARY KEY, FOREIGN KEY, and CHECK constraints in the Orders and Customers tables.
- Queries on Relational Database using GROUP BY, HAVING and ORDER BY clauses of SQL.

MODULE-2

UNIT- I

6L+6T+6SL+6P=24 Hours

NORMALIZATION:

Complex Queries, Triggers, Views: More complex SQL retrieval queries; Views in PL/SQL.

Basics of Functional Dependencies and Normalization for Relational Databases: Informal design guidelines for relation schemas; Functional dependencies-inference rules, equivalence and minimal cover; Normal forms based on primary keys; Boyce-Codd normal form; Properties of relational decompositions, multi valued dependency, join dependencies.

UNIT- II

6L+6T+6SL+6P=24 Hours

INTRODUCTION TO TRANSACTION PROCESSING CONCEPTS AND CONCURRENCY CONTROL TECHNIQUES:

Introduction to transaction processing; Transaction and system concepts; Desirable properties of transactions; Characterizing schedules based on serializability, Two-phase locking techniques for concurrency control.

UNIT- III

6L+6T+6SL+6P=24 Hours

DATABASE RECOVERY TECHNIQUES: Recovery concepts; Shadow paging; The ARIES recovery algorithm:

INDEXING STRUCTURES FOR FILES AND PHYSICAL DATABASE DESIGN: Single level and multi-Level indexing; Dynamic multi-level indexing using B-trees and B+ trees.

PRACTICES:

- Design and Development of company database and expressing Nested queries using SQL.
- Design and Development of student database and specifying queries using set operations.
- Design and Development of sailor's database and specifying queries using different types of JOINS.
- Creation and dropping of VIEWS.
- Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?
- Apply various DCL and TCL constructs of SQL on Relational Database.

SKILLS:

- ✓ Build databases for web, mobile, and enterprise applications.
- ✓ Extract insights from large datasets using advanced SQL queries.
- ✓ Maintain, secure, and optimize organizational databases.
- ✓ Ensure safe and consistent handling of transactions (ACID properties).
- ✓ Qualify for roles like backend developer, data analyst, database administrator or data engineer

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1.	Apply the fundamental concepts of database systems to identify database users, system components, and architecture.	Apply	1, 1
2.	Analyze real-world applications to identify entities, relationships, and constraints for ER modeling.	Analyze	1, 2
3.	Analyze database schemas to identify functional dependencies and normalize them using appropriate normal forms.	Analyze	2, 1
4.	Design and implement database applications using PL/SQL, views to handle real-world problems.	Create	2, 1
5.	Analyze transaction schedules to determine serializability and identify concurrency issues.	Analyze	2, 2
6.	Evaluate concurrency control technique two-phase locking to ensure database consistency.	Evaluate	2, 3

UNIT-WISE MAPPING OF SDGS

Unit	Topic	Sustainable Development Goals
M – I Unit – I	Databases And Database Users, Database System Concepts and Architecture, Conceptual Data Modeling and Database Design	SDG 4 – Quality Education SDG 9 – Industry, Innovation & Infrastructure
M – I Unit – II	Relational Database Design by ER–To-Relational Mapping, The Relational Data Model and Relational Database Constraints	SDG 4 - Quality Education, SDG 9 – Industry, Innovation & Infrastructure, SDG 12 – Responsible Consumption & Production
M – II Unit – I	Complex Queries, Triggers, Views, Functional Dependencies and Normalization for Relational Databases	SDG 9 - Industry, Innovation & Infrastructure, SDG 12 - Responsible Consumption & Production
M – II Unit – II	Transaction Processing Concepts and Theory	SDG 9 – Industry, Innovation & Infrastructure, SDG 16 – Peace, Justice, and Strong Institutions
M – II Unit – III	Concurrency Control Techniques, Database Recovery Techniques	SDG 9 – Industry, Innovation & Infrastructure, SDG 11 – Sustainable Cities & Communities

MAPPING OF SDGS AND IKS

Module–Unit	Topic	IKS Theme
M–I, Unit–1	Databases, DBMS characteristics, users, system architecture	Grantha Samraksana
M–I, Unit–1	Schemas, instances, three-schema architecture, data independence	Layered Vedic Knowledge Structure
M–I, Unit–	ER modeling: entities, attributes, relationships, weak entities	Ayurvedic & Vedanga Taxonomies; Upaveda as Weak Entities
M–I, Unit–2	Relational model, keys, constraints	Gotra, Nama, Desa; Temple and Kingdom Records
M–I, Unit–2	ER-to-Relational mapping, SQL basics	Catalogs of Palm-Leaf Manuscript Collections
M–II, Unit–1	Functional dependencies, normalization, BCNF, MVD, JD	Nyaya & Logical Consistency in Indian Logic
M–II, Unit–2	Transactions, serializability, schedules	Temple Treasury & Royal Administrative Procedures
M–II, Unit–3	Concurrency control, recovery, ARIES, indexing	Preservation & Restoration of Inscriptions/Manuscripts

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):**TEXTBOOKS:**

1. Ramez, Elmasri and Shamkant B. Navathe, "Fundamentals of Database Systems", 7th Edition, Pearson Education, 2016.
2. Raghu Rama Krishnan and Johannes Gehrke, "Database Management Systems", 3rd Edition, Tata McGraw Hill, 2013.

REFERENCE BOOKS:

1. Abraham Silberschatz, Henry F.Korth and S.Sudarshan, "Database System Concepts", 7th edition, Tata Mc Graw Hill, 2019.
2. Allen G. Taylor "Database Development for Dummies" 1st Edition, 2011
3. C. J. Date "Introduction to Database Systems" 7th Edition, Addison Wesley, 2003.